



Arash Pashazadeh

Curriculum Vitae

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Date of birth: 16th March 1990

Education

- **Ph.D. Student, Systems Engineering**
Dept. of Civil, Coastal, and Environmental Engineering, University of South Alabama, Mobile, AL, United States, Spring 2025, Supervisor: Dr. Shenghua Wu
- **M.Sc., Civil – Water Engineering and Hydraulic Structure**
Dept. of Civil Engineering, Razi University, Kermanshah, Iran, 2015 - 2018
Thesis: Development of Hydrologic Models for Flow Routing in Multiple-Branch Systems (Evaluated and approved by thesis committee: Excellent, 19.25 out of 20), Supervisor: Dr. Mitra Javan
- **B.Sc., Civil Engineering**
Dept. of Civil Engineering, Kurdistan University, Kurdistan, Iran, 2010 –2014
- **HSE-MS certificate**
HSE-MS certificate Cert No: 137IR0519 (www.asysac.ir) - 2018

Articles

- Pashazadeh, A., Javan, M. (2019). Comparison of the Gene Expression Programming, Artificial Neural Network (ANN) and Equivalent Muskingum Inflow models in the Flood Routing of Multiple-Branched Rivers, *Theoretical and Applied Climatology journal*. Quartile 2 (Q2). Cited: 31 <https://doi.org/10.1007/s00704-019-03032-2>
- Pashazadeh, A., Javan, M. (2019). Improvement of lumped models in multiple inflows rivers by principal component analysis and reliability analysis. *Acta Geophysica*, 1-13. Quartile 2 (Q2) Cited: 4 <https://doi.org/10.1007/s11600-019-00290-6>
- Pashazadeh, A., Javan, M., Wu, S. (2025). Development of Nonlinear Muskingum Models in Multiple Inflows River Systems. <https://doi.org/10.21203/rs.3.rs-703757/v1>
- Pashazadeh, A., Wu, S. (2025). Issues in the Effect of Applied Explicit Numerical Solutions on the Accuracy of Nonlinear Muskingum Models. [Ready to Submit](#).
- Song, J., Pashazadeh, A., Wu, S. (2025). Advances in Microplastic Detection and Interception: A State-of-the-Art Review. [Under Review](#).



Google Scholar profile: <https://scholar.google.com/citations?user=kjy-7YkAAAAJ&hl=en>

Instagram profile: (Visual description of my projects): <https://www.instagram.com/arashcivilengineering/>

Research and Work Interests

#Numerical hydrologic and hydraulic modeling #Flood forecasting #Stormwater management #Porous Media
#Machine learning #Artificial Intelligence programming #Aquatic and non-aquatic structures design

Language Skills

English: Total TOEFL IBT test score 104 out of 120 /Reading: 26/30, Listening:28/30, Speaking: 21/30 and Writing 29/30

Honors and Recommendation Letters

- ✓ Ranked 30th nationwide in Iran (2018) in the Ph.D. candidate selection exam for Water Resources Management Engineering, evaluated by the National Organization of Educational Testing (NOET).
- ✓ Recommendation Letters:
 - University of Western Sydney, Sydney, Australia - Prof (Full) Ataur Rahman
 - Accepted to Morgan State University, Maryland, USA, Prof. Zhuping Sheng
 - Razi University, Kermanshah, Iran Prof. (Full) Houshang Ghamarnia
 - Razi University, Kermanshah, Associate Prof. Mitra Javan

Positions and Memberships

- **Designer & Technical Office Engineer, TTЭМ (TGEM) Construction & Hydropower Company — Dushanbe, Tajikistan (Full-time, 2024)**

Key Project Activities:

- **2024, Shokhdara River, Sebzor: Powerhouse Design**
 - ✓ Designed monorail hoist systems on multiple intake shafts to facilitate equipment handling and maintenance.
 - ✓ Designed Concrete Ferro Beam structures for the powerhouse, ensuring structural stability and load distribution under seismic loads.
 - ✓ Designed the MIV steel System of the powerhouse for safe and efficient water intake.
 - ✓ Designed the steel structure holders for HVAC system installation within the powerhouse.
 - ✓ Designed the thrust rings around the tailrace pipes and thrust ring manifold, providing additional support for high-pressure flow zones.
 - ✓ Structural design of concrete Shafts #1, #2, and #3, for the drainage system of the weir with a depth of 23m.
 - ✓ Designed truss roof structure and crane girder and rail systems of the Powerhouse.
- **2024, Shokhdara River, Sebzor: Headwork Design**
 - ✓ Updating the design of the fishpass-intake structure to allow safe fish migration while managing intake flows.
 - ✓ Developed the design of gabion walls on the right side of the weir to prevent erosion and protect structural integrity.
- **2024, Shokhdara River, Sebzor: Penstock Design**
 - ✓ Designed manhole concrete structure for three penstock sections, providing safe access for inspection and maintenance.
- ✓ **2024, Shokhdara River, Sebzor: Permanent Bridge Design**
 - ✓ Designed steel holder structure for two permanent bridge ends, ensuring stability and load-bearing capacity.
- **2024, Shokhdara River, Sebzor: Headwork**
 - ✓ Prepared sample designs and material specifications for metal works, standardizing fabrication and installation procedures.
- **Design Engineer, Abvarzan Consulting Engineers — Tehran, Iran (Part-time 2018–2020; Full-time 2020- 2024)**

Key Projects:

 - ✓ 2022, Ahvaz: Hydraulic modeling of pump stations; design of suction ponds (1400 m²), stop logs, sluice gates, and industrial steel structures (250–3000 m²).
 - ✓ 2021, Ahvaz: Designed concrete structures for pump stations (2500 m²), control rooms (Ghadir project), 3D modeling of 15,000 m² of landscapes and buildings using SketchUp.
 - ✓ 2020, Andimeshk: Diversion dam design (spillway, ogee, sediment pond), concrete and steel structures for pump stations and warehouses (250–300 m²), suction ponds, control rooms (Balarood project).
 - ✓ 2020, Tehran: Water supply installation project — Three buried concrete tanks (360, 216, 360 m³) and PRK elevated tank (height 20 m).
 - ✓ 2018, Tehran: Control room design of Kordan Dam, FRP reinforcement design.
- **Design Engineer, Arman Pei Alborz Consulting Engineers — Tehran, Iran (Part-time, 2014-2020)**

Key Projects:

 - ✓ 2018, Tehran: Designed the Steel structure of a 2000 m² sports club.
 - ✓ 2017, Tehran: Designed Water supply and sewage networks, street network design for environmental relocation projects in Pakdasht.
 - ✓ 2016, Tehran / Noorabad-Delfan: Designed Water supply and sewage networks for the 6th phase of Parand urban planning projects (80,000–90,000 m²).
 - ✓ 2015, Zanjan: Designed a 16-floor residential concrete building.
 - ✓ 2014, Noorabad-Delfan: Member of design team for 1200 residential units, 80,000 m² urban planning project.
 - ✓ Various years: Design of reinforced concrete retaining walls, pipeline valves, and concrete chambers.
- **Member, Tehran Construction Engineering Organization (TCEO), Membership No. 1030142105**
- **Member, Iran Value Engineering Association, Membership No. 381452**

Digital Skills (Based on work and research experience)

Field	Software
Water Engineering	Flow 3D, EPANET, Water/Sewer CAD
Programming Languages	Python, MATLAB (coding and implementing machine learning and optimization algorithms)
Structural and Road Engineering	CSI Software Suite (SAP2000, SAFE, ETABS), Civil 3D, ABAQUS, PROKON
Geotechnical Engineering	GeoStudio (SEEP/W, SLOPE/W, SIGMA/W, QUAKE/W), PLAXIS
Intelligent Systems	GeneXproTools, NeuroSolutions
Statistical Software	SPSS
Drawing & Design Software	AutoCAD, SketchUp
General Software	Microsoft Project, MS Office (Excel, Word, PowerPoint, Publisher), Photoshop